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Citizenship: United States of America and México

POSITIONS **Seize — Institute of Consulting and Business Solutions** 2025 - *present*
Economic Modeling Advisor
Riyadh, Saudi Arabia
Develop and deploy computable general equilibrium (CGE) and gravity models to assess trade agreements and policy reforms.
Build institutional modeling capacity, including AI-powered natural-language interfaces that let stakeholders query model results and run scenarios without writing code.
Pair data-driven and economics-driven analysis to inform policy decisions, and deliver capacity-building workshops for government staff.

Independent Consulting 2025
Economic Modeling Consultant
Chicago, IL

USDA Economic Research Service 2022 - 2025
Research Agricultural Economist
Market and Trade Economics Division,
Agricultural and Policy Models branch

EDUCATION **Ph.D., Agricultural and Applied Economics** 2021
University of Illinois Urbana-Champaign, IL
Advisor: William Ridley
Dissertation: Three Essays on Agricultural and Consumer Economics, *Outstanding Doctoral Dissertation Award*
Committee: William Ridley (chair), Sandy Dall’Erba, Yilan Xu, and Rafael Garduño-Rivera

B.A., Mathematical Economics 2016
California State University, San Bernardino, San Bernardino, CA
Advisor: Kazim Konyar
Summa Cum Laude

RESEARCH INTERESTS	Methods	Fields
	Bayesian data analysis	Agricultural production
	General equilibrium models	Consumption economics
	Structural econometrics	Food systems
	Machine learning & AI for policy	

JOURNAL ARTICLES 1. *Land competition and poverty effects from the Mexico’s proposal to ban Genetically Engineered corn.* 2024 **American Journal of Agricultural**

- Economics** 106(3): 1300-1325.
with Jayson Beckman, Angelica Williams, and Steven Zahniser.
doi:10.1111/ajae.12463
2. *Losing Ground: Measuring the Welfare Effects of Retail Food Price Inflation during the COVID-19 Pandemic on Mexican Household.* 2024 **Journal of Agriculture and Food Industrial Organization** 22 (2): 143-154.
with Benjamin D.K. Wood, and Rafael Garduño-Rivera.
doi:10.1515/jafio-2023-0041
 3. *Estimation of the Value-Added/Intermediate Input Substitution Elasticities Consistent with the GTAP Data.* 2023. **Journal of Global Economic Analysis** 8(2): 134-158.
with Jayson Beckman, and Maros Ivanic.
doi:10.21642/JGEA.080204AF
 4. *A model of the U.S. food system: What are the determinants of the state vulnerabilities to production shocks and supply chain disruptions?* 2023. **Agricultural Economics** 54(1): 95-109. ¹ ²
with William Ridley, and Sandy Dall’Erba.
doi:10.1111/agec.12750
 5. *The impact of taxing sugary-sweetened beverages in México: A censored QUA demand system approach.* 2022. **Journal of Agricultural and Applied Economics Association** 1(1): 18-32.
with Diansheng Dong.
doi:10.1002/jaa2.6
 6. *U.S. interstate trade will mitigate the negative impact of climate change on crop profit.* 2021. **American Journal of Agricultural Economics** 103(5): 1720-1741.
with Sandy Dall’Erba, and Zhangliang Chen.
doi:10.1111/ajae.12204
 7. *The impacts of agroforestry on agricultural productivity, ecosystem services, and human well-being in low- and middle-income countries: An evidence and gap map.* 2020. **Campbell Systematic Reviews** 16(1): e1066.
with Daniel C. Miller, Pablo J. Ordoñez, Sarah E. Brown, Samantha Forrest, Karl Hughes, and Kathy Baylis.
doi:10.1002/cl2.1066
 8. *The conduits and barriers to reentry for formerly incarcerated individuals in San Bernardino.* 2018. **Journal of Prison Education and Reentry** 5(1): 2-17.
with Annika Anderson, and Patricia Cortez.
doi:10.25771/sdf0-1631

¹Most Outstanding Second Year Research Paper at University of Illinois at Urbana-Champaign-ACE 2020.

²The Charles M. Tiebout Prize in Regional Science 2021.

- BOOK CHAPTERS
1. *Climate and Weather Impacts on Agricultural Productivity*. 2025. **Encyclopedia of Energy, Natural Resource, and Environmental Economics**, 2e. with Eric Njuki, and Boris E Bravo-Ureta.
- U.S.
DEPARTMENT OF
AGRICULTURE,
ECONOMICS
RESEARCH
SERVICE
1. *International Income and Price Elasticity Estimates: An Update* 2025 **Technical Bulletin** TB-1971. with Jacob Abrehe Zereyesus, Tian Xia, Xianghong Li, and Lila Cardell
 2. *Potential Budgetary Impacts of Climate Change on the Pasture, Rangeland, and Forage Insurance Plan* 2024 **Economic Brief** EB-38: 25pp. with Dylan Turner, Katherine Baldwin, Jayson Beckman, Francis Tsiboe, and Kate Vaiknoras.
 3. *ERS Research Models Future Effects of Climate Change on Corn and Soybean Yields, Production, and Exports* 2024 **Amber Waves** 1545-8741. with Jayson Beckman. ERS Amber Waves, March 04, 2024
 4. *Estimating the market and trade implications from climate-Induced corn and soybean yield changes for the U.S.* 2023 **Economics Research Report** ERR-324: 31pp. with Jayson Beckman, and Maros Ivanic.
 5. *Warming temperatures in U.S. Corn Belt expected to continue into next decade.* Charts of Note, December 5, 2023
- WORKING
PAPERS
1. *Zeros, Elasticities, and the Soda Tax: Estimator Choice in Censored Demand Systems.* Solo paper. Under review at the **American Journal of Agricultural Economics**.
 2. *Inflation and the Cost of Delayed Indexation in México's Soda Tax.*
 3. *Tracing U.S. county-to-county domestic trade flows in grains.* with William Ridley, Sandy Dall'Erba, Jian Zou, Youning Chen, and Christa Court. AAEE discussion paper
 4. *The Impact of International Demand for Cereals on U.S. Crop Land Appreciation: A Model Implied Instrumental Variable Approach* with William Ridley, and Stephen Morgan. AAEE discussion paper
 5. *Does trade openness affect global grain supply response?* with Jacob Zereyesus, Jennifer Kee, Stephen Morgan, Nathan Hendricks, Michael Johnson, Lila Cardell, Jeremy Jelliffe, and Walter Acpangan. AAEE discussion paper

6. *Heterogenous patterns of crop yield growth stagnation across U.S. counties in the next decade*
Solo paper.
7. *Revisiting the impact of climate change on agriculture through spatially-varying and place-tailored Ricardian estimates.*
with Sandy Dall’Erba, Chang Cai, and A. Stewart Fotheringham.
CREATE discussion paper

PAPERS IN
PREPARATION

1. *Heterogeneity in Mexican beverage demand: A censored QUAIDS atlas across demographics and survey waves, 2012–2024.*
Solo paper.
2. *Is agricultural insurance a safety net or a net of complacency amidst adaptation to extreme weather events?*
with Francis Tsiboe, Dylan Turner, Katherine Baldwin, and Sharon Raszap.

SOFTWARE
& PUBLIC TOOLS**R & Julia packages**

gravityGE — one-sector Armington-CES gravity model with general equilibrium effects (CRAN; Julia port *GravityGE.jl* on JuliaHub).
censoredAIDS — maximum-likelihood estimation of censored (Quadratic) Almost Ideal Demand Systems (CRAN).

Interactive dashboards

U.S. Domestic Agricultural Trade — interstate agricultural trade flows with gravity / general-equilibrium policy simulation.
 Fertilizer Flows — visualization of global fertilizer trade flows.

AI tools

AI Paper Referee — large-language-model auditor that reviews applied-economics writing against published standards.

FELLOWSHIPS,
GRANTS AND
SCHOLARSHIPS

- The impact of climate change on agricultural production, productivity, and trade (\$403,152) 2023
- Improving the capabilities of global trade models to analyze changes in land use/land cover and green house emissions (\$280,000) 2023
- A global agricultural supply-demand-climate-trade model calibrated to Baseline (\$375,000) 2022
- Towards a data hub of global daily weather forecasts for the ERS (**PI**) (\$30,000) 2022
- ERS’ international supply forecasting (\$30,000) 2022
- International food consumption patterns-implications for global food security and trade (\$25,000) 2022
- The Charles M. Tiebout Prize in Regional Science (\$1,000) 2021
- Cleo Fitzsimmons Award (\$500) 2020
- Agricultural and Consumer Economics Department Grant (\$400) 2019
- Marshall Scott Fellowship (\$4,877.85) 2018
- Brockson Fellowship (\$3,902.28) 2017

	• The Proactive Recruitment in Introductory Science and Mathematics Travel Grant (\$1,000) 2016 • The Proactive Recruitment in Introductory Science and Mathematics Research Grant (\$2,000) 2015 • The John & June Kennedy Economics Scholarship (\$1,500) 2015
AWARDS AND HONORS	• Outstanding Doctoral Dissertation Award, University of Illinois at Urbana-Champaign-ACE 2021 • The Charles M. Tiebout Prize in Regional Science 2021 • Farm foundation Agricultural Scholar 2021 • Most Outstanding Second Year Research Paper, University of Illinois at Urbana-Champaign-ACE 2020
TEACHING AND MENTORSHIP EXPERIENCE	Mentor at ERS Summer internship Kristiina Ala-Kokko (Kansas State University) 2022 Teaching Assistant ACE 240: Personal Financial Planning FA19 Instructor: Paul B Stoddard, Senior Lecturer <i>Lecture:</i> 160 students University of Illinois at Urbana-Champaign, IL Undergraduate Credit: 4 Hours. ACE 231: Food and Agribusiness Management SP18 Instructor: Paul B Stoddard, Senior Lecturer <i>Lecture:</i> 160 students University of Illinois at Urbana-Champaign, IL Undergraduate Credit: 4 Hours. ACE 341AMM: Issues in Agribusiness Markets and Management FA17 Instructor: Paul B Stoddard, Senior Lecturer <i>Lecture:</i> 20 students University of Illinois at Urbana-Champaign, IL Undergraduate Credit: 3 Hours.
RESEARCH IN THE NEWS	1. <i>Climate Impacts on Corn and Soybean Yields by Mid-2030s</i> in Progressive Farmer 2. <i>“Climate change is real;” Researchers warn extreme weather events could impact Illinois agriculture</i> in WCIA.com 3. <i>Trade Insurance</i> in Jayson Lusk, Food and Agricultural Economist, personal blog.
RESEARCH ASSISTANTSHIPS	Interstate Trade as Mitigating Factor for the Effects of Climate Change on Illinois Agriculture

January 2020 - December 2021

How do the interstate trade flows of Illinois farm products respond to extreme weather events?

University of Illinois at Urbana-Champaign, IL

Supervisor: Prof. William Clifton Ridley

WORK

EXPERIENCE

Intern

Department of Agriculture, U.S., Economic Research Service

Summer 2021

Market & Trade Economics Division
Agricultural Policy and Models Branch
Kansas City, MO

Intern

Department of Agriculture, U.S., National Agricultural Statistics Service

2016 - 2017

Environmental, Economics and Demographics
Washington, DC

Intern

Department of Agriculture, U.S., Rural Development
Rural Housing Service
Washington, DC

2015 - 2016

SELECTED

PRESENTATIONS

The Impact of International Demand for Agricultural Products on U.S. Cropland Appreciation: A Model-Implied Instrumental Variable Approach

Agricultural & Applied Economics Association (AAEA) 2024

The Impact of International Demand for Agricultural Products on U.S. Cropland Appreciation: A Model-Implied Instrumental Variable Approach

The International Agricultural Trade Research Consortium (IATRC) 2023
Heterogeneous Patterns of Crop Yield Growth Stagnation across U.S. Stagnation across U.S. Counties in the Next Decade

Agricultural & Applied Economics Association (AAEA) 2023

Track Session: Studying Climate Change Impacts from
Different Methodological Perspectives

The probabilistic role of climate change on crop yield potential

FCSM Research & Policy Conference (FCSM) 2022

The impact of taxing sugar-sweetened beverages in México: A censored QUAID demand system

Agricultural & Applied Economics Association (AAEA) 2022

Track Session: Identification in Demand Estimations

A model of U.S. domestic crop trade to study state's food vulnerability to supply chain disruptions³

Agricultural & Applied Economics Association (AAEA) 2021

Western Regional Science Association (WRSA) 2021

³Previously titled *Who benefits from local agriculture?*

	North American Regional Science Council (NARSC)	2020
	Revisiting the impact of climate change on agriculture through spatially-varying and place-tailored Ricardian estimates	
	North American Regional Science Council (NARSC)	2021
	Workshop RED in Mexico: Challenges of a New Era	2018
REFERENCES	William Clifton Ridley, Ph.D. Professor, Department of Agricultural and Consumer Economics University of Illinois Urbana-Champaign, IL wridley@illinois.edu	
	Sandy Dall’Erba, Ph.D. Professor, Department of Agricultural and Consumer Economics University of Illinois Urbana-Champaign, IL dallerba@illinois.edu	
	Rafael Garduño-Rivera, Ph.D. Professor, School of Economic and Administrative Sciences Universidad Panamericana, México rgardunor@up.edu.mx	
PROGRAMMING & DEPLOYMENT	Julia, L ^A T _E X, Markdown, Python, R, Stan, Stata, & SQL. Docker, AWS, Azure, Shiny Server, Git & GitHub Actions (CI/CD). LLM/AI tooling for natural-language model interfaces and research workflows.	
LANGUAGES	English & Spanish.	
SERVICE AND PROFESSIONAL ACTIVITIES	Journal Refereeing American Journal of Agricultural Economics; Economic Modelling; Discover Food;s Environmental Research Letters; Journal of Agricultural and Applied Economics Association; Nature Food; Organisation for Economic Co-Operation and Development (OECD); U.S. Global Change Research Program (USGCRP); World Development. USDA - Economics Research Service Member, ERS Modeling Team Committee Member, ERS Brownbag Series Committee Abstract Reviewer AAEA Annual Meeting, 2025. AAEA Annual Meeting, 2024.	